Table 5.2 Differences between glucose, fructose and galactose

S. No	Character	Glucose	Fructose	Galactose
1.	Optical rotation	Dexotro	Laevo	Dextro
2.	Solubility in water	Soluble in water	Soluble in water	Slightly soluble
3.	Reducing nature	Reducing sugar	Reducing sugar	Reducing sugar
4.	Action with Bromine water	Oxidised to gluconicacid	Not oxidised	Oxidised to galactonic acid
5.	Action with HNO_3	Oxidised to glucaric acid	Oxidised to a mixture of glycolic, tartaric and trihydroxy glutaric acid	Oxidised to mucic acid
6.	Action with Na/Hg	Reduced to sorbitol	Reduced to a mixture of sorbitol and mannitol	Reduced to dulcitol
7.	Osazone formation	Yellow coloured needle shaped crystals are formed within 10 minutes	Yellow coloured needle shaped crystals are formed within 5 minutes	Yellow coloured broad strick shaped crystals are formed after 20 minutes
8.	Mutarotation	Exhibits	Exhibits	Exhibits
9.	Nature	aldohexose	ketohexose	Aldohexose

5.3 Differences between sucrose, lactose and maltose

S. No	Character	Sucrose	Lactose	Maltose
1.	Commercial Name	Cane sugar	Milk sugar	Maltsugar
2.	Composition	(Glucose + Fructose)	(Galactose + Glucose)	(Glucose + Glucose)
3.	Glycosidic linkage	α(1-2)	α(1-4)	α(1-4)
4.	Reducing Nature	Non reducing Sugar	Reducing Sugar	Reducing Sugar
5.	Hydrolysing Enzyme	Invertase	Lactase	Maltase
6.	Osazone formation	Does not	Yellow colour osazone shaped crystals are formed	Yellow colour tennis ball shaped crystals are formed after 40 minutes

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